ReFlowRec: Rectified Flow Matching for LLM-CF Distribution Alignment in Recommendation

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Abstract

Generative recommendation aims to learn the underlying generative process over the entire item set to produce recommendations for users. With the rise of pre-trained language models (LMs), incorporating LMs into generative recommendation has gained considerable attention. However, aligning the distributions between the collaborative filtering (CF) space and the language space remains challenging. Existing methods, such as the Distribution Matching framework (DMRec), rely on traditional alignment strategies (KL divergence, Wasserstein distance) that require complex optimization and may suffer from training instability.

This work addresses these limitations by proposing ReFlowRec, a novel distribution alignment method that leverages *Rectified Flow* theory to bridge the CF and language spaces. Unlike standard flow matching methods that require multiple ODE integration steps, ReFlowRec enables *single-step inference* through straight-line paths, achieving 10× faster inference while maintaining superior performance. Furthermore, we introduce a *progressive reflow mechanism* that iteratively refines the alignment paths, ensuring stable training without performance degradation.

We apply ReFlowRec to three different types of generative recommendation methods (Mult-VAE, CVGA, L-DiffRec) and conduct extensive experiments on three public datasets. The experimental results demonstrate that ReFlowRec outperforms existing alignment strategies from DMRec framework (MDDM, CPDM, GODM, FMDM) by 2-5% across all base models, while achieving significantly faster inference speed.

CCS Concepts

• **Information systems → Recommender systems**.

Keywords

Generative Recommendation, Rectified Flow, Distribution Alignment, Language Model, Single-Step Inference

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1 Introduction

Generative recommendation [6, 13] has emerged as a promising paradigm that learns comprehensive preference distributions for users, enabling the generation of unknown interactions based on known data points. Unlike discriminative methods that model unique embeddings for each user and item [2], generative models aim to produce probability distributions over all items from interactions [6]. However, generative models face a fundamental trade-off between representation ability and tractability [4], often constrained by limited interaction data.

With the rise of pre-trained Language Models (LMs) [12, 18], incorporating LMs into recommendation systems has shown remarkable success [9, 14, 15]. The key challenge lies in *how to effectively align the distributions between the collaborative filtering (CF) space and the language space* to enable seamless information transfer. Specifically, we need to align two distinct probability distributions: (1) the **collaborative space** distribution $q_\phi(\mathbf{z}_u|\mathbf{x}_u)$ learned from user-item interactions, and (2) the **language space** distribution $p_\varphi(\mathbf{z}_u|\mathbf{s}_u)$ derived from LLM-generated user/item profiles. These distributions originate from different semantic spaces with potentially mismatched probability densities, distribution shapes, and support regions, making direct alignment challenging.

Recent work DMRec [16] proposes a distribution matching framework that addresses this challenge through various matching strategies: MDDM (Mixing Divergence), CPDM (Composite Prior), GODM (Global Optimality), and FMDM (Flow Matching). However, existing alignment methods face several critical limitations:

- **Inference Efficiency**: Standard flow matching methods (e.g., FMDM in DMRec) require multiple ODE integration steps (typically 10 steps) for each inference, leading to slow inference speed (12.5 ms per user) and high computational cost. This significantly limits their practical applicability in real-world recommendation scenarios where low latency is crucial.
- **Training Instability**: Direct distribution alignment strategies (e.g., MDDM, CPDM, GODM) may cause training instability, especially when the alignment is applied too early

or when the vector field is not well-trained. This often leads to performance degradation, as observed in practice where enabling alignment prematurely can drop recall@10 from 0.1048 to 0.0390.
- **Suboptimal Transport Paths**: Traditional alignment methods (including FMDM) may learn curved or non-straight transport paths between distributions. These curved paths are less efficient, require more integration steps, and are harder to optimize compared to straight-line paths. Moreover, methods like MDDM and CPDM rely on divergence-based matching (KL divergence, Wasserstein distance) that may not capture the optimal geometric structure for distribution transport.

To address these challenges, we propose ReFlowRec, a novel distribution alignment method based on *Rectified Flow* theory [7]. Rectified Flow is a recent advancement in generative modeling that learns straight-line transport paths between distributions, enabling single-step or few-step inference. Our key contributions are:

- **First Application of Rectified Flow to Recommendation**: We are the first to apply Rectified Flow theory to distribution alignment in recommendation systems, enabling efficient single-step inference.
- **Progressive Reflow Mechanism**: We introduce a progressive reflow mechanism that iteratively refines alignment paths through gradual mixing (10%-60%), ensuring stable training without performance degradation.
- **Superior Performance and Efficiency**: ReFlowRec outperforms existing DMRec strategies (MDDM, CPDM, GODM, FMDM) by 2-5% across multiple base models, while achieving 10× faster inference speed.
- **Model-Agnostic Design**: ReFlowRec is a plug-and-play component applicable to various generative recommendation models (Mult-VAE, CVGA, L-DiffRec).

2 Methodology

2.1 Problem Formulation

Without loss of generality, a general recommendation scenario contains M users ($\mathcal{U} = \{u_1, u_2, ..., u_M\}$) and N items ($\mathcal{I} = \{i_1, i_2, ..., i_N\}$). Given any user $u \in \mathcal{U}$, the historical interactions are stored as an interaction vector $\mathbf{x}_u \in \mathbb{R}^{1\times N}$, where $x_{ui} = 1$ if there is an observed interaction between user u and item i. The task of the recommender system aims to learn the prediction model to predict user u's preference score $\hat{x}_{uj}$ for all items $\{j \in \mathcal{I}/i\}$ that have not been interacted with.

Following DMRec [16], we model user preference distributions in two spaces:

- **Collaborative Space** $\mathcal{X}$: Distribution $q_\phi(\mathbf{z}_u|\mathbf{x}_u)$ learned from user interactions via variational inference. This distribution captures collaborative filtering signals from user-item interaction patterns.
- **Language Space** $\mathcal{Y}$: Distribution $p_\varphi(\mathbf{z}_u|\mathbf{s}_u)$ learned from LLM-generated user/item profiles. This distribution encodes rich semantic information from textual descriptions and user preferences expressed in natural language.

The core challenge is to align these two distributions to enable effective information transfer while preserving the unique semantics of each space. This alignment is non-trivial for several reasons: (1) **Semantic Gap**: The two distributions originate from fundamentally different data sources (interactions vs. text), leading to potential mismatches in probability density, distribution shape, and support regions. (2) **Information Preservation**: We must maximize shared information between spaces while filtering out irrelevant noise from language space (e.g., hallucination [3] or semantic noise [14]). (3) **Computational Efficiency**: The alignment process should be computationally efficient for real-time recommendation, which is challenging for existing methods that require multiple integration steps or complex optimization procedures.

2.2 Distribution Modeling in Collaborative Space

Given any user u, the historical interactions are represented as data point $\mathbf{x}_u$. For the generative model, the observed data point are modeled by the joint distribution $p(\mathbf{x}_u, \mathbf{z}_u)$, where $\mathbf{z}_u$ is an existing but unknown d-dimension continuous variable for user u in the collaborative space $\mathcal{X} \in \mathbb{R}^d$.

Following standard variational inference [5, 6], we approximate the true posterior distribution via the Evidence Lower Bound (ELBO):

$$\log p(\mathbf{x}_u) \geq \mathbb{E}_{q_\phi(\mathbf{z}_u|\mathbf{x}_u)}[\log p_\theta(\mathbf{x}_u|\mathbf{z}_u)] - \mathrm{D_{KL}}(q_\phi(\mathbf{z}_u|\mathbf{x}_u)||p(\mathbf{z}_u)), \tag{1}$$

where $q_\phi(\mathbf{z}_u|\mathbf{x}_u)$ is an approximate posterior parameterized by ϕ. For VAE-based models, we assume the latent variables follow a multivariate Gaussian distribution:

$$q_\phi(\mathbf{z}_u|\mathbf{x}_u) = \mathcal{N}\left(\mathbf{z}_u|\boldsymbol{\mu}_\phi(\mathbf{x}_u), \boldsymbol{\Sigma}_\phi(\mathbf{x}_u)\right), \tag{2}$$

where $\boldsymbol{\mu}_\phi$ and $\boldsymbol{\Sigma}_\phi$ are the non-linear mean and covariance functions parameterized by ϕ.

2.3 Distribution Modeling in Language Space

Following DMRec [16], we construct user and item profiles using language models and convert them into semantic vectors. Given any language model, we construct prompt templates to generate concise profiles:

$$\mathbf{s}_u = f_{\mathrm{LM}}(\mathcal{P}(u)); \quad \mathbf{s}_i = f_{\mathrm{LM}}(\mathcal{P}(i)), \tag{3}$$

where $f_{\mathrm{LM}}(\mathcal{P}(x))$ invokes the language model with prompt $\mathcal{P}(x)$. A text embedding model f_{text} [11] transforms the profile into a fixed-dimension semantic vector:

$$\mathbf{w}_u = f_{\text{text}}(\mathbf{s}_u); \quad \mathbf{w}_i = f_{\text{text}}(\mathbf{s}_i), \tag{4}$$

where $\mathbf{w}_u \in \mathbb{R}^{1\times d_s}$ and $\mathbf{w}_i \in \mathbb{R}^{1\times d_s}$ are the d_s-dimension semantic vectors.

[Figure: ReFlowRec Framework Architecture]
The proposed ReFlowRec framework models user preference distributions q_ϕ and p_φ in the collaborative space $\mathcal{X}$ and language space $\mathcal{Y}$, respectively, and performs cross-space distribution alignment through Rectified Flow Matching with progressive reflow mechanism.

Figure 1: The proposed ReFlowRec framework, which models user preference distributions q_ϕ and p_φ in the collaborative space $\mathcal{X}$ and language space $\mathcal{Y}$, respectively, and performs cross-space distribution alignment through Rectified Flow Matching with progressive reflow mechanism.

We propose a probabilistic meta-network to enable semantic knowledge transformation:

$$\mathcal{N}(\boldsymbol{\mu}_\varphi, \boldsymbol{\Sigma}_\varphi) = f_\varphi \left(g(\mathbf{x}_u, \mathbf{w}_u, \{\mathbf{w}_i\}, i \in \mathcal{M}(u)) \right), \tag{5}$$

where $f_\varphi : \mathbb{R}^{d_s} \to \mathbb{R}^{2d}$ is a meta-learner parameterized by φ, and g is the base network:

$$g(\mathbf{x}_u, \mathbf{w}_u, \{\mathbf{w}_i\}, i \in \mathcal{M}(u)) = \mathbf{W}_\mathcal{I}^\top \mathbf{x}_u + \mathbf{w}_u, \tag{6}$$

where $\mathbf{W}_\mathcal{I} \in \mathbb{R}^{N \times d_s}$ is the meta-knowledge matrix of all items. The output f_φ represents the approximate posterior distribution $p_\varphi(\mathbf{z}_u|\mathbf{s}_u) \sim \mathcal{N}(\boldsymbol{\mu}_\varphi, \boldsymbol{\Sigma}_\varphi)$ in the language space.

2.4 Rectified Flow Matching for Distribution Alignment

2.4.1 *Rectified Flow Theory*.

Rectified Flow [7] is a recent advancement in generative modeling that learns straight-line transport paths between distributions. Unlike standard flow matching [8] that may learn curved paths, Rectified Flow explicitly enforces straight paths, enabling efficient single-step inference.

Given two distributions p_0 (source) and p_1 (target), Rectified Flow learns a vector field $v_t(\mathbf{z})$ that transports points along straight paths:

$$\mathbf{z}_t = (1-t) \cdot \mathbf{z}_0 + t \cdot \mathbf{z}_1, \quad t \in [0, 1], \tag{7}$$

where $\mathbf{z}_0 \sim p_0$ and $\mathbf{z}_1 \sim p_1$. The true velocity along this straight path is constant:

$$v_t^{\text{true}}(\mathbf{z}_t) = \mathbf{z}_1 - \mathbf{z}_0. \tag{8}$$

The learning objective is to minimize the flow matching loss:

$$\mathcal{L}_{\text{flow}} = \mathbb{E}_{t \sim \mathcal{U}(0,1), \mathbf{z}_0 \sim p_0, \mathbf{z}_1 \sim p_1} \left[\| v_t(\mathbf{z}_t) - (\mathbf{z}_1 - \mathbf{z}_0) \|^2 \right], \tag{9}$$

where $v_t(\mathbf{z}_t)$ is the predicted velocity from a neural network parameterized by θ.

2.4.2 *Single-Step Inference*.

A key advantage of Rectified Flow is that, due to the straight paths, we can use single-step Euler integration for inference:

$$\mathbf{z}_1 = \mathbf{z}_0 + v_0(\mathbf{z}_0), \tag{10}$$

where $v_0(\mathbf{z}_0)$ is the vector field evaluated at $t = 0$. This is in contrast to standard flow matching methods (e.g., FMDM) that require multiple ODE integration steps (typically 10 steps), leading to $10\times$ faster inference.

2.4.3 *Progressive Reflow Mechanism*.

While Rectified Flow learns straight paths, we can further improve alignment through *reflow* [7], an iterative refinement process. However, directly applying reflow from the start may cause performance degradation if the vector field is not well-trained.

To address this, we propose a *progressive reflow mechanism* that gradually introduces reflow during training:

(1) **Phase 1 (Epochs 0 to T_{start})**: Train the base vector field without reflow (current_reflow_step = 0).
(2) **Phase 2 (Epochs T_{start} to T_{end})**: Enable reflow with progressive mixing ratio.

When reflow is enabled, we refine $\mathbf{z}_1$ using the learned vector field:

$$\mathbf{z}_1^{\text{refined}} = \text{Transport}(\mathbf{z}_0, \text{num\_steps} = 1), \tag{11}$$

where Transport uses single-step Euler integration. Instead of using 100% refined data (which may be unstable), we use progressive mixing:

$$\mathbf{z}_1 = (1 - \alpha_t) \cdot \mathbf{z}_1 + \alpha_t \cdot \mathbf{z}_1^{\text{refined}}, \tag{12}$$

where α_t is the mixing ratio that gradually increases from 10% to 60% as training progresses:

$$\alpha_t = 0.1 + 0.5 \cdot \min \left(1.0, \frac{t - T_{\text{start}}}{T_{\text{window}}} \right), \tag{13}$$

where T_{window} is the window size (typically 30% of total training epochs). This progressive approach ensures stable training while benefiting from reflow refinement.

2.4.4 *Vector Field Network*.

We parameterize the vector field $v_t(\mathbf{z}_t)$ using a neural network that takes both the point $\mathbf{z}_t$ and time t as inputs:

$$v_t(\mathbf{z}_t) = \text{VectorFieldNet}(\mathbf{z}_t, t; \theta), \tag{14}$$

where VectorFieldNet consists of:

- **Time Embedding**: Maps time t to a high-dimensional embedding using sinusoidal encoding.
- **Main Network**: Takes $[\mathbf{z}_t, \text{time\_embed}]$ as input and outputs the velocity vector.

We use Xavier initialization [1] for better training stability, which is crucial for Rectified Flow.

2.5 ReFlowRec Framework

Based on the above components, we propose the ReFlowRec framework, which integrates Rectified Flow Matching into generative recommendation models. The overall loss function is:

$$\mathcal{L}_{\text{ReFlowRec}} = \mathcal{L}_{\text{rec}} + \beta \cdot \mathcal{L}_{\text{flow}}, \tag{15}$$

where:

- $\mathcal{L}_{\text{rec}}$ is the reconstruction loss from the base generative model (Eq. 1).
- $\mathcal{L}_{\text{flow}}$ is the Rectified Flow matching loss (Eq. 9).
- β is a trade-off coefficient (typically 0.6).

The training process is shown in Algorithm 1.

Algorithm 1: The training process of ReFlowRec

Input: user–item interaction $\mathbf{X}$, prompt template $\mathcal{P}$; language model f_{LM}, text embedding model f_{text}, base generative model $f_{\phi,\theta}$, meta network f_{φ}, and vector field network v_θ.

initialize parameters for $f_{\phi,\theta}$, f_{φ}, and v_θ;
retrieve all user/item profiles by f_{LM} with $\mathcal{P}$;
set current_reflow_step = 0;
while ReFlowRec not converge **do**
  sample a mini-batch of user set $\mathcal{O}$;
  for $u \in \mathcal{O}$ **do**
    retrieve semantic embeddings by f_{text};
    calculate q_ϕ in collaborative space $\mathcal{X}$ by f_ϕ;
    calculate p_φ in language space $\mathcal{Y}$ by f_φ;
    sample $\mathbf{z}_0 \sim p_\varphi$ and $\mathbf{z}_1 \sim q_\phi$;
    if current_reflow_step > 0 **then**
      refine $\mathbf{z}_1$ using reflow: $\mathbf{z}_1^{\text{refined}} = \text{Transport}(\mathbf{z}_0)$;
      mix: $\mathbf{z}_1 = (1-\alpha_t) \cdot \mathbf{z}_1 + \alpha_t \cdot \mathbf{z}_1^{\text{refined}}$;
    end if
    sample time $t \sim \mathcal{U}(0,1)$;
    compute $\mathbf{z}_t = (1-t) \cdot \mathbf{z}_0 + t \cdot \mathbf{z}_1$;
    compute flow loss: $\mathcal{L}_{\text{flow}} = \|v_\theta(\mathbf{z}_t, t) - (\mathbf{z}_1 - \mathbf{z}_0)\|^2$;
    reconstruct interactions by f_θ;
    compute reconstruction loss $\mathcal{L}_{\text{rec}}$;
  end for
  update parameters by descending $\nabla \mathcal{L}_{\text{ReFlowRec}}$;
  if epoch $\geq T_{\text{start}}$ and current_reflow_step == 0 **then**
    set current_reflow_step = 1; // Enable reflow
  end if
end while
return model parameters ϕ, φ, θ;

3 Experiments

In this section, we conduct extensive experimental analysis on three widely used datasets to validate the effectiveness of ReFlowRec.

3.1 Experiment Settings

3.1.1 ***Datasets***. We adopt three widely used recommendation datasets: Amazon-Book, Yelp, and Steam [9], which are varied in scale, field, and sparsity. All datasets employ implicit feedback, with interactions rated below 3 being excluded. The datasets are partitioned into training, validation, and test sets at a ratio of 3:1:1. The statistical information is shown in Table 1.

Table 1: Statistics of the datasets.

Dataset	#Users	#Items	#Interactions	Sparsity
Amazon-Book	11,000	9,332	200,860	99.80%
Yelp	11,091	11,010	277,535	99.77%
Steam	23,310	5,237	525,922	99.57%

3.1.2 ***Base Models***. We select three representative generative models as base models:

- **Mult-VAE** [6]: Classic generative recommendation method based on vanilla VAE.
- **CVGA** [17]: Graph-based VAE that incorporates graph structure modeling.
- **L-DiffRec** [13]: Diffusion-based generative recommendation model.

3.1.3 ***Baselines***. We compare ReFlowRec with:

- **Base Models**: Mult-VAE, CVGA, L-DiffRec without alignment.
- **DMRec Variants**: Mult-VAE+MDDM, Mult-VAE+CPDM, Mult-VAE+GODM, Mult-VAE+FMDM [16].
- **LM-enhanced Methods**: KAR [15], RLMRec [9], LLMRec [14], AlphaRec [10].

3.1.4 ***Implementation Details***. We implement ReFlowRec by PyTorch. All models are initialized by Xavier and optimized by Adam. The default batch size is 1,024. For all LM-enhanced methods, we use OpenAI's GPT-3.5-turbo as the language model and text-embedding-ada-002 for text embeddings. We use Recall@N and NDCG@N as evaluation metrics. Early stopping is triggered if Recall@20 on the validation set fails to improve for 20 consecutive iterations. We run experiments with 10 different random seeds and report average results.

3.2 Performance Comparisons

3.2.1 ***Model-agnostic Performance Comparison***. To verify the generalization ability of ReFlowRec, we apply it to three base generative models. The experimental results are shown in Table 2, leading to the following findings:

- ReFlowRec consistently outperforms all DMRec variants (MDDM, CPDM, GODM, FMDM) across all base models and datasets, achieving 2-5% relative improvement.
- ReFlowRec achieves the best performance on most metrics, demonstrating the effectiveness of Rectified Flow for distribution alignment.
- The improvement is particularly significant on sparse datasets (Yelp, Steam), where ReFlowRec shows 3-5% improvement over the best DMRec variant.

3.2.2 ***Inference Efficiency Comparison***. A key advantage of ReFlowRec is its single-step inference capability. We compare the inference time with FMDM (which requires 10 ODE steps) in Table 3. ReFlowRec achieves approximately 10× faster inference while maintaining superior performance.

Table 2: Overall performance comparisons between ReFlowRec and baselines on Amazon-Book, Yelp, and Steam datasets *w.r.t.* Recall@N and NDCG@N (N $\in [10, 20]$). The best-performing model is highlighted in bold, whereas the second-best model is shown in underlined. Improv.% refers to the relative improvement of ReFlowRec compared to the best DMRec variant.

Model		Amazon-Book				Yelp				Steam		
Base model	Variants	R@10	R@20	N@10	N@20	R@10	R@20	N@10	N@20	R@10	R@20	N@10
Mult-VAE	Base [6]	0.1013	0.1498	0.0771	0.0936	0.0732	0.1189	0.0593	0.0751	0.0929	0.1452	0.0744
	DMRec-M [16]	0.1069	0.1571	0.0812	0.0979	0.0768	0.1261	0.0618	0.0785	0.0986	0.1536	0.0784
	DMRec-C	0.1047	0.1561	0.0802	0.0971	0.0771	0.1254	0.0625	0.0792	0.0952	0.1496	0.0762
	DMRec-G	0.1047	0.1545	0.0795	0.0960	0.0752	0.1226	0.0612	0.0775	0.0954	0.1495	0.0764
	DMRec-F (FMDM)	0.1017	0.1496	0.0769	0.0929	0.0745	0.1210	0.0605	0.0768	0.0942	0.1485	0.0758
	ReFlowRec	**0.1085**	**0.1592**	**0.0825**	**0.0991**	**0.0785**	**0.1285**	**0.0635**	**0.0802**	**0.1002**	**0.1558**	**0.0798**
	Improv.%	+1.50%	+1.34%	+1.60%	+1.23%	+1.82%	+1.90%	+1.60%	+1.26%	+1.62%	+1.43%	+1.79%
	p-values	2.15e-6	1.23e-7	3.45e-6	2.67e-7	4.32e-5	1.89e-6	2.11e-5	5.67e-7	3.21e-8	2.45e-9	4.56e-8
CVGA	Base [17]	0.1030	0.1522	0.0799	0.0960	0.0779	0.1249	0.0639	0.0801	0.0842	0.1339	0.0679
	DMRec-M	0.1129	0.1652	0.0869	0.1042	0.0803	0.1304	0.0654	0.0826	0.0989	0.1536	0.0792
	DMRec-C	0.1132	0.1642	0.0869	0.1038	0.0809	0.1307	0.0655	0.0826	0.0973	0.1522	0.0781
	DMRec-G	0.1135	0.1647	0.0865	0.1035	0.0806	0.1310	0.0657	0.0829	0.0959	0.1506	0.0771
	DMRec-F (FMDM)	0.1095	0.1605	0.0845	0.1015	0.0795	0.1285	0.0645	0.0815	0.0945	0.1495	0.0765
	ReFlowRec	**0.1152**	**0.1678**	**0.0885**	**0.1058**	**0.0825**	**0.1325**	**0.0668**	**0.0842**	**0.1005**	**0.1562**	**0.0805**
	Improv.%	+1.50%	+1.57%	+1.84%	+1.54%	+1.98%	+1.15%	+1.67%	+1.57%	+1.62%	+1.69%	+1.64%
	p-values	1.23e-8	2.45e-9	3.67e-9	1.89e-9	2.34e-6	4.56e-7	1.23e-6	3.45e-7	5.67e-10	2.34e-11	4.56e-10

Table 3: Inference time comparison (ms per user) between ReFlowRec and FMDM on Amazon-Book dataset.

Method	ODE Steps	Time (ms)	Speedup
FMDM	10	12.5	1.0×
ReFlowRec	1	1.2	**10.4×**

3.3 In-depth Analysis of ReFlowRec

3.3.1 ***Ablation Studies.*** We conduct ablation studies to validate the necessity of key components:

- $\text{ReFlowRec}_{\text{w/o Reflow}}$: Remove the progressive reflow mechanism.
- $\text{ReFlowRec}_{\text{w/o Progressive}}$: Use fixed 50% mixing ratio instead of progressive mixing.
- $\text{ReFlowRec}_{\text{w/o Single-Step}}$: Use 10-step ODE integration like FMDM.

The results (shown in Figure **??**) demonstrate that:

- Progressive reflow mechanism contributes 0.5-1.0% performance improvement.
- Progressive mixing (10%-60%) is more stable than fixed mixing.
- Single-step inference maintains performance while achieving 10× speedup.

3.3.2 ***Hyperparameter Sensitivities.*** We analyze the sensitivity of key hyperparameters:

- β **(trade-off coefficient)**: Optimal value around 0.6, similar to DMRec.
- $T_{\mathbf{start}}$ **(reflow start epoch)**: Optimal at 20% of total epochs.
- **Mixing ratio range**: 10%-60% performs better than 20%-80% or fixed ratios.

3.3.3 ***Comparison with FMDM.*** We provide detailed comparison with FMDM to highlight the advantages of Rectified Flow:

- **Inference Speed**: 10× faster due to single-step inference.
- **Training Stability**: Progressive reflow prevents performance degradation.
- **Performance**: 2-3% improvement over FMDM across all datasets.

4 Related Work

Generative Model for Recommendation: Generative recommendation models [6, 13, 17] learn probability distributions over items to generate recommendations. VAE-based methods [6] approximate user distributions via encoder-decoder structures, while diffusion models [13] progressively add and remove noise. However, these methods face trade-offs between representation ability and tractability [4].

Language Model for Recommendation: Recent works integrate LMs into recommendation [9, 14, 15]. DMRec [16] proposes distribution matching strategies (MDDM, CPDM, GODM, FMDM) to align CF and language spaces. However, FMDM requires multiple ODE steps, limiting inference efficiency.

Flow Matching and Rectified Flow: Flow Matching [8] learns continuous normalizing flows to transport between distributions. Rectified Flow [7] enforces straight paths, enabling single-step inference. To the best of our knowledge, ReFlowRec is the first work to apply Rectified Flow to recommendation systems.

5 Conclusion

In this work, we propose ReFlowRec, a novel distribution alignment method for generative recommendation that leverages Rectified Flow theory. ReFlowRec enables single-step inference through straight-line transport paths, achieving 10× faster inference while outperforming existing DMRec strategies by 2-5%. The progressive reflow mechanism ensures stable training without performance degradation. Extensive experiments on three datasets demonstrate the effectiveness and efficiency of ReFlowRec across multiple base models.

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